Abbas Taqvi

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Professional Summary

- Data engineer and M.S. Data Science student at Milano-Bicocca with experience building cloud data platforms and batch and streaming pipelines. Built an AWS lakehouse for querying 30B+ records in under a second, a Rust ETL framework that processes data 10× faster than the legacy .NET system, and a LangGraph Text2SQL agent that reduced analyst turnaround by 41%. Seeking data engineering internship opportunities in Italy.

Education

- **Università degli Studi di Milano-Bicocca** — Milan, Italy
  Master of Science in Data Science; Grade: 26/30 — *Oct 2025 – Present*
- **Habib University** — Karachi, Pakistan
  Bachelor of Science in Computer Science; GPA: 3.31; 100% merit scholarship (HU-TOPS) — *Aug 2020 – May 2024*

Experience

- **Logiciel Services LLC** — United States (Remote)
  Data Engineer I — *May 2024 – Nov 2025*
  - **Data Platform**: Built the data platform from scratch, using Airflow for orchestration, Spark (Scala) for processing (ETL/ELT), Kafka for streaming, Amazon S3 for storage, Iceberg as the table format, Trino for query engine and AWS Glue for metadata. Simplified a 12-component architecture to 5, housing 30B+ records queryable in <1s.
  - **Ferroflow**: Rust ETL framework with composable processor/pipeline abstractions, replacing a legacy .NET monolith: 500 min → 50 min daily (10× speedup), saving 400 EC2-minutes per day.
  - **Text2SQL Chatbot**: Multi-agent chatbot (Python, LangChain, LangGraph, Trino) that translates natural language into SQL that queries the data lake via REST API, reducing analyst turnaround by 41% for business teams.
- **ImproData** — Pinedale, WY (Remote)
  Junior Data Engineer — *Jan 2021 – May 2022*
  - **Data Pipelines**: Built pipelines on Azure Databricks (PySpark) and automated ingestion via GCP Cloud Functions (JSON → GCS → BigQuery), eliminating 3 hours of daily manual work.
- **Pullflow Inc.** — San Jose, CA (Remote)
  DevOps Intern — *Jul 2023 – Aug 2023*
  - **Kubernetes Migration**: Migrated production services from AWS EKS to on-premise Kubernetes (MicroK8s): wrote Dockerfiles, Helm charts and a Traefik reverse-proxy config, with a zero-downtime cutover.
- **United Bank Limited** — Karachi, Pakistan
  Data Engineering Intern — *May 2023 – Jun 2023*
  - **Migration Pipeline**: Built a cross-region migration pipeline in Informatica and validated ingestion across 3 countries.

Projects

- **Kuza: End-to-End Data Platform**: Spark (Scala) ingestion for 375+ daily sessions, Kafka streaming, S3 lakehouse, Trino and Airflow with DAGs auto-generated from YAML (Scala, Spark, Kafka, Airflow, Trino, AWS, Docker).
- **Yohsin: Zero-Copy Object Serializer**: Rust serializer using memory-mapped files, O(1) random access and a write-ahead log for sub-50 ms recovery; sustained 1M records/s (2.25 GB/s). Built in 24 hours at a hackathon (Rust, tokio, mmap).

Skills

- **Languages**: Python, Rust, Scala, C/C++ (CUDA), Go, SQL, JavaScript
- **Data & Cloud**: Spark, Kafka, Airflow, Trino, Iceberg, Databricks, AWS (S3, EKS, EC2, Glue), Azure, GCP
- **Systems & Infrastructure**: Linux, Docker, Kubernetes (EKS, MicroK8s), Traefik, REST APIs, gRPC, Git
- **AI / ML**: PyTorch, JAX, TensorFlow, LangChain, LangGraph, Pydantic-AI
- **Databases**: PostgreSQL, MongoDB, Neo4j, Qdrant, Weaviate, Amazon RDS
- **Spoken Languages**: English (C1), Italian (A2)

Achievements & Test Scores

- **Tests Scores**: IELTS 8.0/9.0; GRE Quantitative 165/170
- **Competitive Programming**: ICPC Asia Topi Regionals 2022: 19th place; Google Kickstart (May 2022): 6th place country-wide

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